

Data Summarization

Data Wrangling in R

Data Summarization

- Basic statistical summarization
 - `mean(x)`: takes the mean of x
 - `sd(x)`: takes the standard deviation of x
 - `median(x)`: takes the median of x
 - `quantile(x)`: displays sample quantiles of x. Default is min, IQR, max
 - `range(x)`: displays the range. Same as `c(min(x), max(x))`
 - `sum(x)`: sum of x
 - `max(x)`: maximum value in x
 - `min(x)`: minimum value in x
- all have the `na.rm` = argument for missing data

Statistical summarization

These functions work on **vectors**:

```
x <- c(1, 5, 7, 4, 2, 8)  
mean(x)
```

```
[1] 4.5
```

Summarization on a `data.frame`/`tibble`:

```
mtcars %>% pull(hp) %>% mean()
```

```
[1] 146.6875
```

GUT CHECK!

What kind of object do we need to run summary operators like `mean()` ?

- A. A vector of numbers
- B. A vector of characters
- C. A dataset

UFO Data

Let's use the UFO data again:

```
ufo <- read_csv("https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_data_complete.csv")
ufo <- ufo %>% rename(duration_s = `duration (seconds)`)
head(ufo)
```

```
# A tibble: 6 × 11
  datetime city state country shape duration_s `duration (hours/min)` comments
  <chr>    <chr> <chr> <chr>   <chr>      <dbl> <chr>                        <chr>
1 10/10/19... san ... tx    us    cyli...    2700 45 minutes                This ev...
2 10/10/19... lack... tx    <NA>    light    7200 1-2 hrs                  1949 La...
3 10/10/19... ches... <NA>    gb     circ...     20 20 seconds                Green/O...
4 10/10/19... edna  tx    us     circ...     20 1/2 hour                  My olde...
5 10/10/19... kane... hi    us     light     900 15 minutes                AS a Ma...
6 10/10/19... bris... tn    us     sphe...     300 5 minutes                My fath...
# i 3 more variables: `date posted` <chr>, latitude <chr>, longitude <chr>
```

Column to vector

Let's work with one column as a vector using `pull()`.

```
ufo_shapes <- ufo %>% pull(shape)
ufo_shapes
```

```
[1] "cylinder" "light"    "circle"   "circle"   "light"    "sphere"
[7] "circle"   "disk"     "disk"     "disk"     "circle"   "fireball"
[13] "disk"     "unknown"  "oval"     "circle"   "disk"     "disk"
[19] "light"    "light"    "other"    "disk"     "light"    "light"
[25] "oval"     "cigar"    "other"    "light"    "rectangle" "chevron"
[31] "triangle" "oval"     "unknown"  "sphere"   "unknown"  "light"
[37] "disk"     "circle"   "other"    "formation" "triangle"  "chevron"
[43] "disk"     "triangle" "unknown"  "light"    "unknown"  "disk"
[49] "triangle" "triangle" "unknown"  "triangle" "unknown"  "triangle"
[55] "formation" "unknown"  "cigar"    "unknown"  "sphere"   "light"
[61] "other"     "cigar"    "rectangle" "light"    "sphere"   NA
[67] NA          NA         "triangle" "light"    "light"    "cylinder"
[73] "delta"     "sphere"   "triangle" "other"    "disk"     "changing"
[79] "fireball"  "cylinder" "cigar"    "circle"   "triangle"  "triangle"
[85] "light"     "egg"      "fireball" "changing" "egg"      "unknown"
[91] "other"     "fireball" "sphere"   "circle"   "light"    "light"
[97] "sphere"    "flash"    "triangle" "chevron"  "oval"     "oval"
[103] "flash"     "other"    "cylinder" "triangle" "oval"     "light"
[109] "light"     "unknown"  "diamond"  "triangle" "formation" "other"
[115] "circle"    "rectangle" "changing" "triangle" "triangle"  "flash"
[121] "unknown"   "circle"   "sphere"   "unknown"  "other"    "circle"
[127] "egg"       "cross"    "light"    "oval"     "sphere"   "other"
[133] "light"     "oval"     "formation" "light"    "unknown"  "sphere"
```

Check for NAs

use `sum(is.na())`:

```
sum(is.na(ufo_shapes))
```

```
[1] 3118
```

dpLyr: count

Use `count` directly on a `data.frame` and `column`: count the number of rows in each group.

```
ufo %>% count(shape)
```

```
# A tibble: 30 × 2
  shape      n
  <chr>    <int>
1 changed      1
2 changing 2140
3 chevron  1007
4 cigar    2241
5 circle   8453
6 cone     367
7 crescent    2
8 cross     265
9 cylinder 1382
10 delta      8
# i 20 more rows
```


dpLyr: count

Multiple columns listed further subdivides the count.

```
ufo %>% count(shape, country)
```

```
# A tibble: 139 × 3
  shape    country     n
  <chr>    <chr>   <int>
1 changed  us         1
2 changing au        10
3 changing ca       73
4 changing de         2
5 changing gb        47
6 changing us     1708
7 changing <NA>     300
8 chevron  au         3
9 chevron  ca        38
10 chevron de         1
# i 129 more rows
```

dp~~l~~yr: count

Option to sort the results with `sort = TRUE`

```
ufo %>% count(shape, sort = TRUE)
```

```
# A tibble: 30 × 2
```

	shape	n
	<chr>	<int>
1	light	17872
2	triangle	8489
3	circle	8453
4	fireball	6562
5	unknown	6319
6	other	6247
7	disk	6005
8	sphere	5755
9	oval	4119
10	<NA>	3118

```
# i 20 more rows
```

GUT CHECK!

The `count()` function can help us tally:

- A. Sample size
- B. Rows per each category
- C. How many categories

dp`lyr`: count

Instead of counting the **number of rows** in each group, `wt` computes `sum()` of a column for each group.

```
# Add up "duration_s" for each "shape" category  
ufo %>% count(shape, wt = duration_s)
```

```
# A tibble: 30 × 2  
  shape      n  
  <chr>    <dbl>  
1 changed    3600  
2 changing 4137860.  
3 chevron   492047.  
4 cigar    4045685.  
5 circle  36407947.  
6 cone    26182820  
7 crescent  37810  
8 cross    174465  
9 cylinder 4400398.  
10 delta   16155  
# i 20 more rows
```

Grouping

Perform Operations By Groups: dplyr

Regular data...

```
# No group_by()
ufo
```

```
# A tibble: 88,875 × 11
```

	datetime	city	state	country	shape	duration_s	`duration (hours/min)`	comments
	<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>	<chr>	<chr>
1	10/10/1...	san ...	tx	us	cylind...	2700	45 minutes	This ev...
2	10/10/1...	lack...	tx	<NA>	light	7200	1-2 hrs	1949 La...
3	10/10/1...	ches...	<NA>	gb	circ...	20	20 seconds	Green/O...
4	10/10/1...	edna	tx	us	circ...	20	1/2 hour	My olde...
5	10/10/1...	kane...	hi	us	light	900	15 minutes	AS a Ma...
6	10/10/1...	bris...	tn	us	sphe...	300	5 minutes	My fath...
7	10/10/1...	pena...	<NA>	gb	circ...	180	about 3 mins	penarth...
8	10/10/1...	norw...	ct	us	disk	1200	20 minutes	A brigh...
9	10/10/1...	pell...	al	us	disk	180	3 minutes	Strobe ...
10	10/10/1...	live...	fl	us	disk	120	several minutes	Saucer ...

```
# i 88,865 more rows
```

```
# i 3 more variables: `date posted` <chr>, latitude <chr>, longitude <chr>
```

Perform Operations By Groups: dplyr

`group_by` allows you group the data set by variables/columns you specify:

```
ufo_grouped <- ufo %>% group_by(shape)
ufo_grouped
```

```
# A tibble: 88,875 × 11
# Groups:   shape [30]
  datetime city state country shape duration_s `duration (hours/min)` comments
  <chr>    <chr> <chr> <chr>   <chr>      <dbl> <chr>                        <chr>
1 10/10/1... san ... tx    us    cyli...    2700 45 minutes                This ev...
2 10/10/1... lack... tx    <NA>   light    7200 1-2 hrs                  1949 La...
3 10/10/1... ches... <NA>   gb    circ...     20 20 seconds                Green/O...
4 10/10/1... edna  tx    us    circ...     20 1/2 hour                  My olde...
5 10/10/1... kane... hi    us    light     900 15 minutes                AS a Ma...
6 10/10/1... bris... tn    us    sphe...     300 5 minutes                 My fath...
7 10/10/1... pena... <NA>   gb    circ...     180 about 3 mins              penarth...
8 10/10/1... norw... ct    us    disk    1200 20 minutes                A brigh...
9 10/10/1... pell... al    us    disk     180 3 minutes                 Strobe ...
10 10/10/1... live... fl    us    disk     120 several minutes         Saucer ...
# i 88,865 more rows
# i 3 more variables: `date posted` <chr>, latitude <chr>, longitude <chr>
```

Summarize the grouped data

We can summarize grouped data using `summarize()`:

```
ufo_grouped %>%  
  summarize(mean_value = mean(duration_s, na.rm = TRUE))
```

```
# A tibble: 30 × 2  
  shape      mean_value  
  <chr>      <dbl>  
1 changed      3600  
2 changing    1934.  
3 chevron      489.  
4 cigar       1805.  
5 circle      4308.  
6 cone       71343.  
7 crescent   18905  
8 cross        658.  
9 cylinder    3184.  
10 delta      2019.  
# i 20 more rows
```


Use the **pipe** to string these together!

Pipe `ufo` into `group_by`, then pipe that into `summarize`:

```
ufo %>%  
  group_by(shape) %>%  
  summarize(mean_value = mean(duration_s, na.rm = TRUE),  
            max_value = max(duration_s, na.rm = TRUE))
```

```
# A tibble: 30 × 3  
  shape      mean_value max_value  
  <chr>      <dbl>      <dbl>  
1 changed      3600         3600  
2 changing    1934.        172800  
3 chevron      489.         43200  
4 cigar      1805.        1814400  
5 circle      4308.       10526400  
6 cone      71343.       25248000  
7 crescent   18905         37800  
8 cross       658.         10800  
9 cylinder   3184.       2631600  
10 delta     2019.        14400  
# i 20 more rows
```

group_by with mutate - Useful for comparisons

Use `group_by` to calculate the mean value for each shape. We can use `mutate` to add it as a column.

```
ufo_compare <- ufo %>%  
  group_by(shape) %>%  
  mutate(duration_shape_avg = mean(duration_s, na.rm = TRUE)) %>%  
  select(shape, duration_s, duration_shape_avg)  
ufo_compare
```

```
# A tibble: 88,875 × 3  
# Groups:   shape [30]  
  shape      duration_s duration_shape_avg  
  <chr>         <dbl>         <dbl>  
1 cylinder      2700           3184.  
2 light        7200          12256.  
3 circle        20           4308.  
4 circle        20           4308.  
5 light         900          12256.  
6 sphere        300          20411.  
7 circle        180           4308.  
8 disk         1200          2018.  
9 disk          180           2018.  
10 disk         120           2018.  
# i 88,865 more rows
```

group_by with mutate - Useful for comparisons

Create a “difference” variable:

```
ufo_compare %>% mutate(diff = duration_s - duration_shape_avg)
```

```
# A tibble: 88,875 × 4
```

```
# Groups:   shape [30]
```

	shape	duration_s	duration_shape_avg	diff
	<chr>	<dbl>	<dbl>	<dbl>
1	cylinder	2700	3184.	-484.
2	light	7200	12256.	-5056.
3	circle	20	4308.	-4288.
4	circle	20	4308.	-4288.
5	light	900	12256.	-11356.
6	sphere	300	20411.	-20111.
7	circle	180	4308.	-4128.
8	disk	1200	2018.	-818.
9	disk	180	2018.	-1838.
10	disk	120	2018.	-1898.

```
# i 88,865 more rows
```

Use `n()` for sample size by group

There are other functions, such as `n()` count the number of observations.

```
ufo %>%  
  group_by(shape) %>%  
  summarize(n = n(),  
            mean = mean(duration_s, na.rm = TRUE))
```

```
# A tibble: 30 × 3  
  shape      n  mean  
  <chr>  <int> <dbl>  
1 changed      1 3600  
2 changing  2140 1934.  
3 chevron   1007  489.  
4 cigar    2241 1805.  
5 circle   8453 4308.  
6 cone      367 71343.  
7 crescent    2 18905  
8 cross      265  658.  
9 cylinder  1382 3184.  
10 delta      8 2019.  
# i 20 more rows
```

Iterative summaries

Iterative summaries: `dplyr` `summarize()` and `across()` functions

Use the `across` function with `summarize()` to summarize across multiple columns of your data.

```
dropouts <- read_delim("https://sisbid.github.io/Data-Wrangling/data/dropouts.txt", delim = "/")
dropouts %>%
  group_by(ETHNIC) %>%
  summarize(across(c(D9, D10, D11, D12), ~sum(.x)))
```

A tibble: 9 × 5

	ETHNIC	D9	D10	D11	D12
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	1226	299	308	478
2	1	103	82	124	265
3	2	118	114	132	767
4	3	37	25	34	135
5	4	34	30	51	252
6	5	4433	3543	4723	16220
7	6	817	617	875	2890
8	7	768	878	1432	3998
9	9	162	155	194	636

Select different columns based on the data class

Use ?tidyr_tidy_select functions like `where(is.numeric)`!

```
dropouts %>%  
  group_by(ETHNIC) %>%  
  summarize(across( where(is.numeric), ~ sum(.x, na.rm = TRUE)))
```

A tibble: 9 × 18

	ETHNIC	E7	E8	E9	E10	E11	E12	EUS	ETOT	D7	D8
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	2922	2709	3140	3291	2776	2345	43	11595	66	52
2	1	2523	2589	2808	2886	2889	3020	33	11636	16	15
3	2	44199	43898	43343	43757	45840	43001	596	176537	69	37
4	3	2390	2284	2436	2352	2471	2571	42	9872	9	7
5	4	12210	12883	13429	14037	14273	14469	233	56441	12	8
6	5	254745	252583	264302	260201	252458	253193	2680	1032834	682	550
7	6	26383	26786	28497	28731	28696	30799	485	117208	217	164
8	7	113498	115139	115593	117592	119308	121771	1829	476093	323	267
9	9	14704	13469	13999	13238	12810	13000	138	53185	59	38

i 7 more variables: D9 <dbl>, D10 <dbl>, D11 <dbl>, D12 <dbl>, DUS <dbl>,
DTOT <dbl>, YEAR <dbl>

summary() Function

Using `summary()` can give you rough snapshots of each numeric column (character columns are skipped):

```
summary(dropouts)
```

CDS_CODE	ETHNIC	GENDER	E7
Length:59599	Min. :0.000	Length:59599	Min. : 0.000
Class :character	1st Qu.:2.000	Class :character	1st Qu.: 0.000
Mode :character	Median :5.000	Mode :character	Median : 0.000
	Mean :4.737		Mean : 7.946
	3rd Qu.:7.000		3rd Qu.: 3.000
	Max. :9.000		Max. :373.000
E8	E9	E10	E11
Min. : 0.000	Min. : 0.00	Min. : 0.000	Min. : 0.000
1st Qu.: 0.000	1st Qu.: 0.00	1st Qu.: 0.000	1st Qu.: 0.000
Median : 0.000	Median : 0.00	Median : 0.000	Median : 0.000
Mean : 7.925	Mean : 8.18	Mean : 8.156	Mean : 8.079
3rd Qu.: 3.000	3rd Qu.: 2.00	3rd Qu.: 2.000	3rd Qu.: 2.000
Max. :363.000	Max. :567.00	Max. :537.000	Max. :620.000
E12	EUS	ETOT	D7
Min. : 0.000	Min. : 0.000	Min. : 0.00	Min. : 0.00000
1st Qu.: 0.000	1st Qu.: 0.000	1st Qu.: 0.00	1st Qu.: 0.00000
Median : 0.000	Median : 0.000	Median : 1.00	Median : 0.00000
Mean : 8.124	Mean : 0.102	Mean : 32.64	Mean : 0.02438
3rd Qu.: 2.000	3rd Qu.: 0.000	3rd Qu.: 9.00	3rd Qu.: 0.00000
Max. :490.000	Max. :77.000	Max. :2127.00	Max. :15.00000
D8	D9	D10	D11
Min. :0.00000	Min. : 0.0000	Min. : 0.00000	Min. : 0.0000

Summary

- summary stats (`mean()`) work with `pull()`
- `count(x)`: what unique values do you have?
- `group_by()`: changes all subsequent functions
 - combine with `summarize()` to get statistics per group
 - combine with `across()` to select several columns
- `summary(x)`: quantile information

<https://sisbid.github.io/Data-Wrangling/labs/data-summarization-lab.Rmd>